2025.12.8 HW10 习题六

B2. $\overline{X} \sim N(\mu, \frac{\sigma^2}{9})$

$\therefore$ (1) $N(0,1)$ (2) $t(8)$ (3) $E(\overline{X}) = E(X) = \mu$.

$\therefore$ 为 $\chi^2(8)$

(4) $\chi^2(9)$ (5) $\chi^2(1)$ (6) $F(1,8)$

(7) $F(1,2)$ (8) $F(2,2)$

B9. $X_1, X_2 \cdots X_8 \sim N(0,1)$

$\overline{X} \sim N(0, \frac{1}{8})$. $7S^2 \sim \chi^2(7)$

$X_9 - \overline{X} \sim N(0, \frac{9}{8})$.

$\therefore Y \sim t(7)$

B10. $\overline{X} \sim N(\mu, \frac{\sigma^2}{5})$, $\overline{Y} \sim N(\mu, \frac{\sigma^2}{9})$.

(1)

$\therefore \frac{a(\overline{X}-\overline{Y})}{\sigma} \sim N(0, \frac{14a^2}{45})$. $\Rightarrow a = \pm\sqrt{\frac{45}{14}}$.

(2) $\frac{(\overline{X}-\overline{Y})}{\sqrt{\frac{4S_1^2+8S_2^2}{12}}\sqrt{\frac{1}{5}+\frac{1}{9}}} \sim t(12) \Rightarrow b = \pm\sqrt{\frac{135}{14}}$

B11. 不妨 $X \sim N(\mu_1, \sigma^2)$, $Y \sim N(\mu_2, \sigma^2)$.

则 $\frac{S_1^2/\sigma^2}{S_2^2/\sigma^2} \sim F(6,6)$.

即 $\frac{S_1^2}{S_2^2} \sim F(6,6)$.

$\therefore C = F_{0.025}(6,6) \approx 5.82$

习题七.

A3. (1) $\hat{\theta}_1 = \frac{\overline{X}}{2-\overline{X}}$, $\overline{X} = \frac{0.45+0.2+0.5+0.47+0.35+1.63+0.14+0.06+0.89+0.34}{10}$

$= 0.503$.

$\therefore \hat{\theta}_1 = 0.336$.

(2) $\hat{\theta}_1 = \sqrt{\frac{1}{2}A_2}$, $A_2 = \frac{1}{10}(0.05^2+0.47^2+0.01^2+0.03^2+0.18^2+1.65^2+0.64^2+1.05^2+0.41^2+0.19^2)$

$= 0.46956$

$\therefore \hat{\theta}_1 = 0.4845$.

(3) $\hat{\theta}_1 = 2\overline{X}-2$, $\overline{X} = 1.093$

$\therefore \hat{\theta}_1 = 0.186$

A5. $E(X) = 2\theta(1-\theta) + 2(1-\theta)^2$

$= 2 - 2\theta$.

$\bar{x} = \frac{2+0+2+1+0+0+1+2+1}{9} = 1$. $\bar{X} = E(X)$

$\Rightarrow 2\hat{\theta} = 2 - \bar{X}$.

$\therefore \hat{\theta}_1 = \frac{1}{2}(2-\bar{x}) = \frac{1}{2}$.

B1. $E(X) = \int_{-\infty}^{+\infty} x f(x;\theta) dx = \int_0^\theta \frac{6}{\theta^3}(\theta x^2 - x^3) dx$

$= \frac{\theta}{2}$.

$\therefore \bar{X} = \frac{\theta}{2}$, $\theta = 2\bar{X}$.

故 $\hat{\theta} = 2\bar{X}$.

$Var(X) = \int_0^\theta x^2 f(x;\theta) dx$

$= \int_0^\theta \frac{6}{\theta^3} x^3(\theta - x) dx$

$= \frac{\theta^2}{20}$

$E(\hat{\theta}) = 2E(\bar{X}) = \theta$

$Var(\hat{\theta}) = 4Var(\bar{X}) = \frac{\theta^2}{5n}$